Hints: Number theory – 3

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. For each x, the number of y with $y^2 = c$ is $1 + \left(\frac{c}{p}\right)$. So $N = p + \sum_x \left(\frac{1-x^2}{p}\right)$.

Hint 2. Key lemma: $\sum_x \left(\frac{ax^2+bx+c}{p}\right) = -\left(\frac{a}{p}\right)$ if $b^2 - 4ac \neq 0$, and $(p-1)\left(\frac{a}{p}\right)$ if $b^2 - 4ac = 0$. Prove it by completing the square and reducing to $\sum_x \left(\frac{x^2-d}{p}\right)$, $d \neq 0$. That sum equals -1: count the solutions of $x^2 - y^2 = d$, i.e. $(x-y)(x+y) = d$, in two ways.

Hint 3. Hence $N = p - \left(\frac{-1}{p}\right)$. For a general conic, complete squares to reach $au^2 + bv^2 = c$ and apply the lemma. A non-degenerate conic has $p - \left(\frac{-ab}{p}\right)$ affine points.

Problem 2

Hint 1. By Möbius inversion, $N_q = \sum_{d|q} \mu(d) \lfloor q/(4d) \rfloor$. For odd e, $\lfloor e/4 \rfloor = \frac{e-2+\chi_4(e)}{4}$, where χ_4 is the non-trivial character mod 4.

Hint 2. For $q > 1$ this gives $4N_q = \varphi(q) + \chi_4(q) \prod_{p|q}(1 - \chi_4(p))$. The product is 0 if some $p \equiv 1 \pmod 4$ divides q, and $2^{\omega(q)}$ otherwise.

Hint 3. N_q is odd if and only if $4N_q \equiv 4 \pmod 8$. Since $2^{\omega(q)} \mid \varphi(q)$, two or more prime factors give an even N_q. For $q = p^k$, check $p \bmod 8$ and the parity of k: $p \equiv 5$ (where $N_q = \varphi(q)/4$) and $p \equiv 7$ give odd N_q, while $p \equiv 1, 3$ give even N_q.

Problem 3

Hint 1. Thue: consider $u - av$ for $0 \leqslant u, v \leqslant \lfloor \sqrt{p} \rfloor$. There are more than p such pairs, so two coincide mod p; subtract. With $a^2 \equiv -1$ this gives $p \mid x^2 + y^2 < 2p$.

Hint 2. For $p \equiv 1 \pmod 3$, -3 is a quadratic residue (reciprocity). Take $a^2 \equiv -3$. Then $p \mid x^2 + 3y^2 < 4p$, so $x^2 + 3y^2 \in \{p, 2p, 3p\}$.

Hint 3. $2p$ is impossible: x, y would have the same parity, giving $4 \mid x^2 + 3y^2$. If it is $3p$, then $3 \mid x$ and $p = y^2 + 3(x/3)^2$.

Problem 4

Hint 1. Squares are $0, 1, 4 \pmod 8$, so three of them never sum to $7 \pmod 8$. If $4 \mid x^2 + y^2 + z^2$, then all of x, y, z are even. Descend.

Hint 2. By Euler's four-square identity it suffices to treat primes. For odd p, the sets $\{x^2\}$ and $\{-1 - y^2\}$ each have $\frac{p+1}{2}$ elements mod p, so they meet: $x^2 + y^2 + 1 \equiv 0$, i.e. mp is a sum of four squares for some $m < p$.

Hint 3. Take m minimal and suppose $m > 1$. Reduce the x_i to $y_i \equiv x_i \pmod m$ with $|y_i| \leqslant m/2$; then $\sum y_i^2 = mr$ with $r < m$. Multiply the two representations using Euler's identity: every term is divisible by m. Dividing by m^2 represents rp, a contradiction. (Treat the case where all $|y_i| = m/2$ separately.)

Problem 5

Hint 1. Use multiplicativity in both arguments. The exponent $\frac{n-1}{2}$ is additive mod 2 on odd numbers: $\frac{ab-1}{2} \equiv \frac{a-1}{2} + \frac{b-1}{2} \pmod 2$. So the reciprocity sign for m, n is the product of the signs for all prime pairs $p \mid m$, $q \mid n$. The supplements follow in the same way.

Hint 2. Example: $(\frac{2}{15}) = (\frac{2}{3})(\frac{2}{5}) = 1$, yet 2 is not a square mod 3, hence not mod 15.

Hint 3. Solovay–Strassen: test whether $a^{(n-1)/2} \equiv (\frac{a}{n}) \pmod n$. The Jacobi symbol is computed fast by reciprocity. For odd composite n the a passing the test form a proper subgroup of $(\mathbb{Z}/n)^*$, so a random a fails with probability $\geqslant \frac{1}{2}$.

Problem 6

Note. The equation is $\sqrt[3]{7x^2 - 13xy + 7y^2} = |x - y| + 1$.

Hint 1. $x = y$ gives $(1, 1)$. Otherwise let $x > y$ and $n = x - y$, and cube both sides. You get $y^2 + ny = n^3 - 4n^2 + 3n + 1$.

Hint 2. Complete the square: $(2y + n)^2 = (n - 2)^2(4n + 1)$. So $4n + 1$ is an odd square, $n = m^2 + m$ with $m \geqslant 2$.

Hint 3. Then $2y + m^2 + m = (m - 1)(2m^2 + 5m + 2)$. Answer: $(1, 1)$ and $\{x, y\} = \{m^3 + 2m^2 - m - 1,\ m^3 + m^2 - 2m - 1\}$, $m \geqslant 2$.

Problem 7

Hint 1. $p = 2$: all $x + y = 2^k$ work. For odd p, let $x^{p-1} + y = p^a$ and $x + y^{p-1} = p^b$ with $x \leqslant y$, so $a \leqslant b$. Substitute $y = p^a - x^{p-1}$ and work mod p^a.

Hint 2. This gives $p^a \mid x^{(p-1)^2} + x$. Show $p \nmid x$, so $p^a \mid x^{p(p-2)} + 1$, and $p \mid x + 1$.

Hint 3. By LTE, $v_p(x^{p(p-2)} + 1) = v_p(x + 1) + 1 =: r + 1$, so $a \leqslant r + 1$, while $p^r \leqslant x + 1 \leqslant p^a$. This forces $x = p - 1$ and $a = 2$. Then $(p - 1)^{p-1} < p^2$ forces $p = 3$. Answer: $(3, 2, 5)$, $(3, 5, 2)$, and $p = 2$ with $x + y$ a power of 2.

Problem 8

Hint 1. Five suffice: $x^2 + 7 = x^2 + 2^2 + 1^2 + 1^2 + 1^2$. With four squares, each $f_i = a_i x + b_i$ is linear and $\sum a_i^2 = 1$, $\sum a_i b_i = 0$, $\sum b_i^2 = 7$.

Hint 2. So b lies in the orthogonal complement of a rational unit vector a in $\mathbb{Q}^4$. By Witt cancellation, the form $\sum x_i^2$ restricted to $a^\perp$ is $\mathbb{Q}$-equivalent to $x^2 + y^2 + z^2$. Hence 7 would be a sum of three rational squares.

Hint 3. Clearing denominators, $7m^2$ would be a sum of three integer squares. But $7m^2$ has the form $4^k(8l + 7)$ (Problem 4). (The official solution does the descent mod 8 by hand.) Answer: $n = 5$.

Problem 9

Hint 1. For $p \equiv 1 \pmod 4$, $(\frac{p-a}{p}) = (\frac{a}{p})$. Pair a with $p - a$: each pair contributes $p(\frac{a}{p})$.

Hint 2. The sum equals $p \sum_{a<p/2} (\frac{a}{p})$. That inner sum is half of $\sum_{a=1}^{p-1} (\frac{a}{p}) = 0$, so the whole sum is 0.

Hint 3. For $p \equiv 3 \pmod 4$, $p > 3$, Dirichlet's class number formula gives $h(-p) = -\frac{1}{p} \sum_{a=1}^{p-1} (\frac{a}{p}) a$. For example, $p = 7$ gives -7, i.e. $h = 1$; $p = 23$ gives -69, i.e. $h = 3$.

Problem 10

Hint 1. $g = \sum_a (\frac{a}{p})\zeta^a$ with $\zeta = e^{2\pi i/p}$. Compute $g\bar{g}$ or $g^2 = \sum_{a,b}(\frac{ab}{p})\zeta^{a+b}$: substitute $b = at$ and sum over a first. You get $g^2 = (\frac{-1}{p})p =: p^*$.

Hint 2. In $\mathbb{Z}[\zeta]$, $g^q \equiv \sum(\frac{a}{p})\zeta^{qa} = (\frac{q}{p})g \pmod{q}$ (Frobenius, using $q \mid \binom{q}{j}$).

Hint 3. Also $g^q = g\,(p^*)^{(q-1)/2} \equiv (\frac{p^*}{q})g$. Multiply by g and cancel the unit $p^* \bmod q$: $(\frac{q}{p}) = (\frac{p^*}{q})$, which is reciprocity.